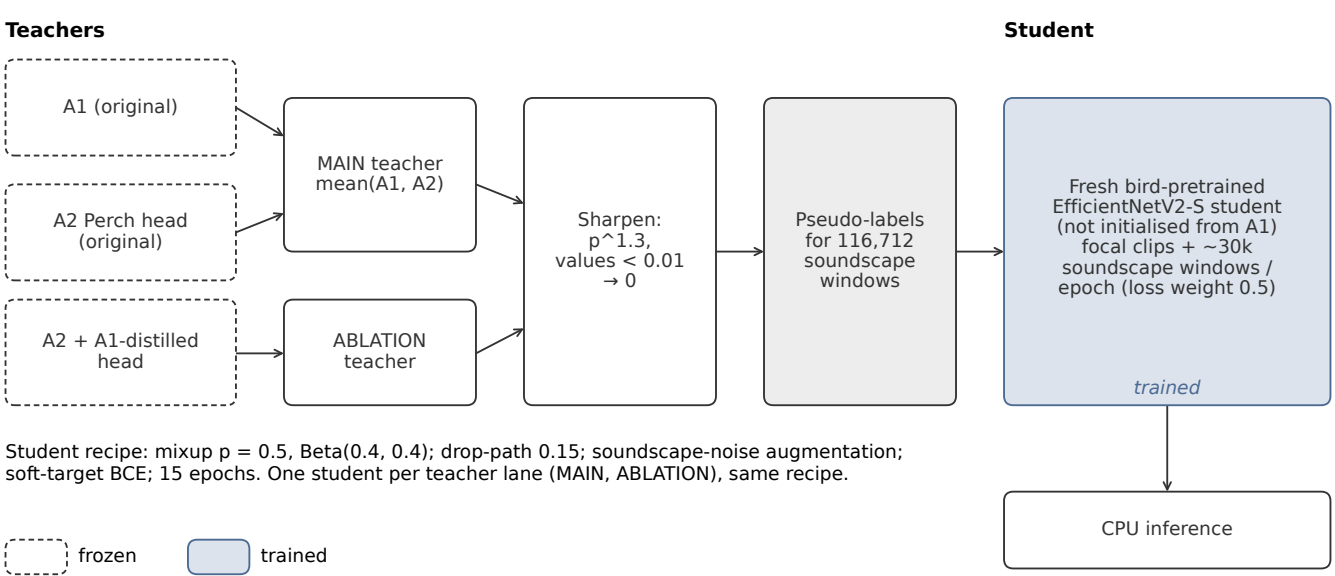
Teachers
Student
A1 (original)
A2 Perch head (original)
A2 + A1-distilled head
MAIN teacher mean(A1, A2)
ABLATION teacher
Sharpen: p^1.3, values < 0.01 → 0
Pseudo-labels for 116,712 soundscape windows
Fresh bird-pretrained EfficientNetV2-S student (not initialised from A1) focal clips + ~30k soundscape windows / epoch (loss weight 0.5)
trained
CPU inference
Student recipe: mixup p = 0.5, Beta(0.4, 0.4); drop-path 0.15; soundscape-noise augmentation; soft-target BCE; 15 epochs. One student per teacher lane (MAIN, ABLATION), same recipe.
frozen
trained